

SolutionRequirements

Date	01 October 2026
TeamID	SWTID-2026-2606
ProjectName	PocketSmart AI: Your Smart Budget&RecommendationAssistant
MaximumMarks	4Marks

DefineFunctionalandNon-FunctionalRequirements

Functional requirements describe what PocketSmart AI should do. Non-functional requirements describe how the systemshouldperform and the quality constraints it should satisfy.

Step1:FunctionalRequirements

S.No	RequirementCategory	RequirementDescription	Priority
1	Authentication	Userscanregister,loginandlogoutusingse ssion-basedauthentication.	High
2	Authorizationlevels	Authenticateduserscanaccesstheir own planner and recommendationhistory; no admin role is required forthe current scope.	Medium
3	Externalinterfaces	The application uses the Gemini APIfor AI recommendations and providesrelevantexternalproduct/servic elinks.	High
4	Transactionsprocessing	Planner inputs are validated andprocessedforrecommendations;thes ystemdoesnotprocesspayments.	Medium
5	Reporting	Thesystemdisplaysrecommendationres ultsandallowsuserstoviewsavedrecom mendationhistory.	Medium
6	Businessrules,etc.	Recommendationsshouldrespecttheuser 's stated budget, category andrequirements.Invalidormissinginputsh ould be handled safely.	High
7	Compliancetolawsorregulations	Protect account and sessioninformation and avoid storingunnecessary sensitive user data. Nospecificregulated-domaincomplian ceisclaimed.	Medium
8	Other	SupportHomeInterior,Party/Eventan d Jewelry planners, includingoptional jewelry image input.	High

Step2:Non-FunctionalRequirements

S.No	NFR	RequirementDescription	Target/Measure
1	Performance&Speed	Planner pages should loadresponsivelyandnormalAPIrequests should complete without timeout.	ResponsiveUI;normalrequestscomplete without timeout.
2	Scalability	Modular FastAPI routes anddatabase-backedhistoryall owincrementalgrowth.	Extendmoduleswithoutredesigningthe core system.
3	Security&DataPrivacy	Passwordsarehashed,sessionsusea secret key, and uploaded imagesare validated and size-limited.	No plaintext passwords; rejectunsupportedoroversizedimages.
4	Reliability&Availability	Fallback/defaultrecommendationsarea available when the AI service isunavailable.	Supportedfallbackscenariosremainusabl e.
5	Usability&Accessibility	Clearforms,readableresults,responsivel ayoutsandsimplenavigation are provided.	Userscancompleteplannerflowswithout technical knowledge.
6	Other	Separateroutes,services,templatesand tests support maintainability.	Modulesremainorganizedandtestable.